

NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte (Deemed to be University)

II Sem B.Tech. (CBCS) Mid Semester Examinations - I, February 2023

ME1003-1 – ELEMENTS OF MECHANICAL ENGINEERING

Duration: 1 Hour

Max. Marks: 20

*Note: Answer any **One** full question from **each Unit**.*

	Unit – I	Marks	BT*	CO*	PO*
1. a)	Explain the difference between conventional and nonconventional energy sources.	5	L*2	1	1
b)	A mixture of saturated water and saturated steam at a temperature of 250°C is contained in a closed vessel of 0.1m ³ capacity. If the mass of the saturated water is 2kg, find the mass of the steam in the vessel. Also find the pressure, specific volume, dryness fraction and the enthalpy of the mixture. From steam tables, At saturation temperature $T_s = 250^\circ\text{C}$ the pressure is $p = 39.77\text{bar}$ and $v_f = 0.0012513\text{m}^3/\text{kg}$ $v_g = 0.05004\text{m}^3/\text{kg}$ $h_f = 1085.8\text{kJ/kg}$ $h_{fg} = 1714.6\text{ kJ/kg}$.	5	L3	1	2
2. a)	Label the major parts of Babcock and Wilcox boiler with a neat sketch and describe the course of flue gas and water circulation in it.	8	L1	1	1
b)	Define the following terms with respect to steam, (i) Saturation temperature. (ii) Latent heat of evaporation.	2	L2	1	2
	Unit – II				
3. a)	Explain the working concept of a centrifugal pump with a neat sketch.	5	L1	2	1
b)	Illustrate the difference between open and closed cycle gas turbine.	5	L2	2	1
4. a)	Define pumps and show the classification of pumps?	5	L1	2	1
b)	Demonstrate with simple illustration the working of a reaction steam turbine.	5	L2	2	1

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
